

Binary Detection

June 16, 2026

1 Class Assignments

1.1 Assignment 2: Effect of Prior Probabilities

Objective. Analyze how unequal prior probabilities modify the MAP decision threshold and the resulting error probabilities.

Problem statement. Use the same binary AWGN model:

$$H_0 : Y = a_0 + \eta, \quad (1)$$

$$H_1 : Y = a_1 + \eta, \quad (2)$$

with

$$\eta \sim \mathcal{N}(0, \sigma_\eta^2). \quad (3)$$

Let the prior probability of H_1 vary as

$$\pi_1 \in \{0.1, 0.3, 0.5, 0.7, 0.9\}, \quad (4)$$

with

$$\pi_0 = 1 - \pi_1. \quad (5)$$

Tasks.

1. For each value of π_1 , compute the MAP threshold γ_{MAP} .
2. For each value of π_1 , compute:

$$P_{\text{FA}}, \quad P_{\text{M}}, \quad P_e. \quad (6)$$

3. Plot γ_{MAP} as a function of π_1 .
4. Plot P_{FA} , P_{M} , and P_e as functions of π_1 .
5. Explain why the MAP threshold moves toward the less probable hypothesis.
6. Discuss why, for sufficiently unequal priors and high noise variance, the MAP threshold may move outside the interval $[a_0, a_1]$.

Required deliverables.

1. Table of π_1 , γ_{MAP} , P_{FA} , P_{M} , and P_e .
2. Plot of γ_{MAP} versus π_1 .
3. Plot of error probabilities versus π_1 .
4. Short interpretation of the role of prior probabilities in MAP detection.